



Mechanics of displacive instabilities in solids

Habilitation à Diriger des Recherches (HDR)

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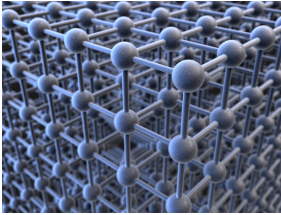
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Non-linear phenomena

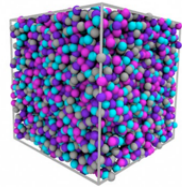
Crystal Transitions and Disorder

Crystalline



Stronger disorder
→
←
Weak disorder

Amorphous

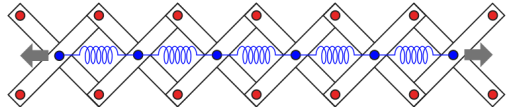


Meta-materials

spring-chain



spring-chain system embedded into a pantograph



Scientific Context: Nonlinearity & Instability

Scientific Challenges

The Challenge:

Understanding material behavior beyond the elastic limit where microscale rearrangements drive macroscopic nonlinearity.

Key Phenomena:

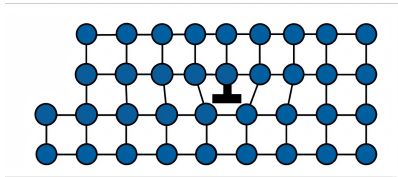
- ▶ **Phase Transitions:** Martensitic transformations.
- ▶ **Plasticity:** Dislocation-mediated flow and avalanches.
- ▶ **Fracture:** Brittle-to-ductile transitions and damage.

Unifying Perspective

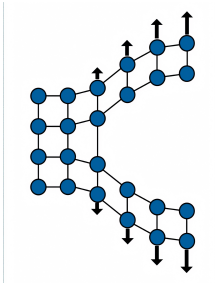
These phenomena are inherently **displacement-driven** (*non-diffusive*) and can be studied within the **same context** of displacive instabilities.

Common Physical Mechanisms: Displacive Instabilities

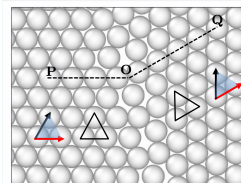
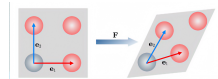
Plasticity



Fracture



Phase Transition

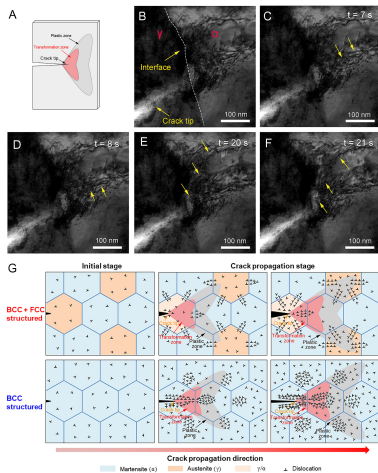


Central Theme

Uniform treatment through the lens of non-convex energy and geometric frustration.

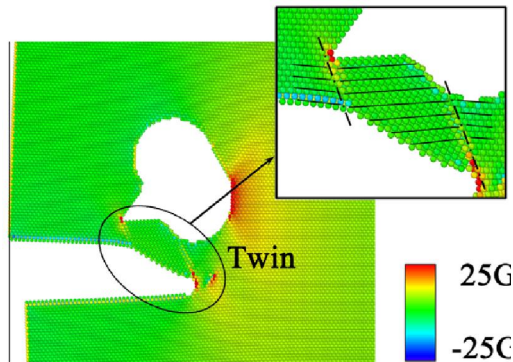
Motivation for Unification

Experimental Observations



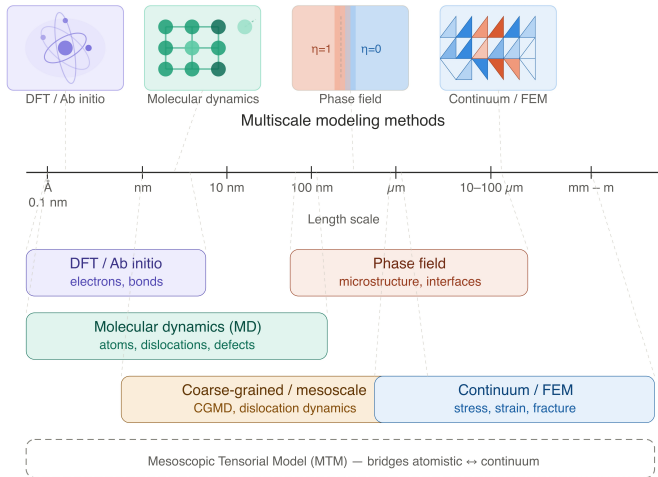
Zhang et al. PNAS, 2025

Computational Modeling

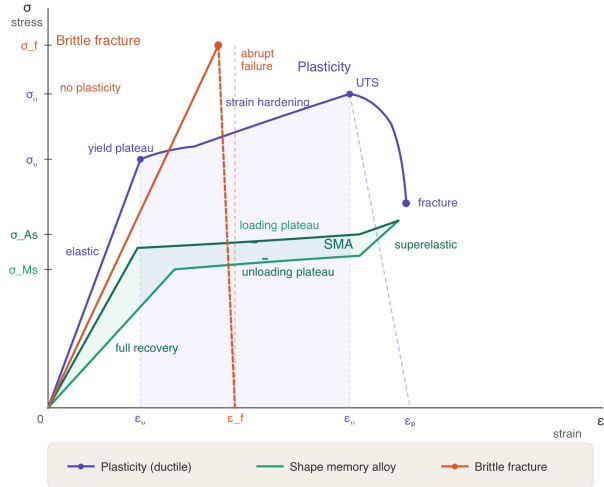


Liu, Groh, Mater. Sci. Eng., 2014

Multiscale Modeling Framework



Macroscopic Response & Engineering Scale



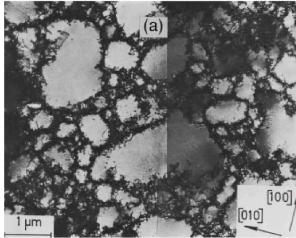
Bridging the gap between material design and structural performance.

Self-Organization and Spatial Patterns

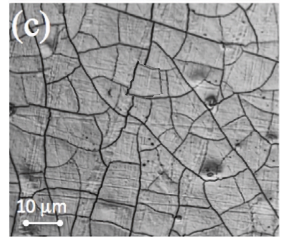
Twinning



Dislocation Cells



Fracture Patterns

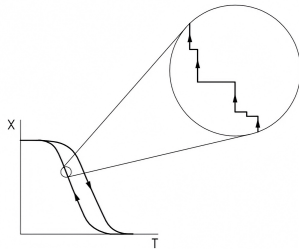


Complexity through Instability

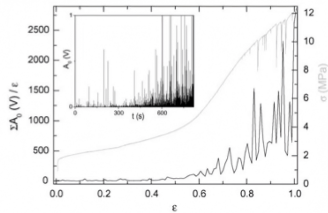
Spatio-temporal organization and collective behavior.

Comparison of Mechanisms

Mechanism A



Mechanism B



Mechanism C

